



清華大學

4.36

$$(a) x(t) = (e^{-t} + e^{-2t})u(t) \leftrightarrow X(j\omega) = \frac{1}{1+j\omega} + \frac{1}{2+j\omega} = \frac{4+j(2\omega)}{(1+j\omega)(2+j\omega)} = 2 \cdot \frac{2+j\omega}{(1+j\omega)(3+j\omega)}$$

$$y(t) = [2e^{-t} - 2e^{-2t}]u(t) \leftrightarrow Y(j\omega) = 2\left(\frac{1}{1+j\omega} - \frac{1}{2+j\omega}\right) = 2 \cdot \frac{3}{(1+j\omega)(4+j\omega)}$$

$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} = \cancel{2} \times \frac{3+j\omega}{(4+j\omega)(2+j\omega)} = \cancel{2} \times \frac{3}{2} \times \left(\frac{1}{4+j\omega} + \frac{1}{2+j\omega}\right)$$

$$(b) h(t) = \frac{3}{2}(e^{-2t} + e^{-4t})u(t)$$

$$(c) y(t) = x(t) * h(t) = \int_{-\infty}^{t+\infty} (e^{-\tau} + e^{-2\tau})u(\tau) \cdot \frac{3}{2}(e^{-2(t-\tau)} + e^{-4(t-\tau)})u(t-\tau) d\tau$$

$$= u(t) \int_0^t (e^{-\tau} + e^{-2\tau})u(\tau) \cdot \frac{3}{2}(e^{-2t+2\tau} + e^{-4t+4\tau})u(t-\tau) d\tau$$

$$= u(t) \int_0^t \frac{3}{2}(e^{-\tau} + e^{-2\tau})(e^{-2t+2\tau} + e^{-4t+4\tau}) d\tau$$

$$= \frac{3}{2}u(t) \left[e^{-2t} \int_0^t (e^{-\tau} + e^{-2\tau}) d\tau + e^{-4t} \int_0^t (e^{-\tau} + e^{-2\tau}) d\tau \right]$$

$$= \frac{3}{2}u(t) \left[e^{-2t} \int_0^t x(\tau) e^{2\tau} d\tau + e^{-4t} \int_0^t x(\tau) e^{4\tau} d\tau \right]$$

$$\frac{dy(t)}{dt} = \frac{3}{2} \left[\frac{d}{dt} \left(e^{-2t} \int_0^t x(\tau) e^{2\tau} d\tau + e^{-4t} \int_0^t x(\tau) e^{4\tau} d\tau \right) + u(t) \left[-2e^{-2t} \int_0^t x(\tau) e^{2\tau} d\tau + x(t) + (-4e^{-4t}) \int_0^t x(\tau) e^{4\tau} d\tau + 4x(t) \right] \right]$$

$$(c) H(j\omega) = \frac{3(j\omega) + 9}{6(j\omega)^2 + (8-j\omega^2)} = \frac{3(j\omega) + 9}{(j\omega)^2 + 6(j\omega) + 8}$$

$$6 \cdot \frac{dy(t)}{dt} + 8y(t) = 3 \frac{dx(t)}{dt} + 9x(t)$$

4.44

$$(a) \int_{-\infty}^{\infty} x(\tau) z(t-\tau) d\tau = u(t) \int_0^t x(\tau) e^{-(t-\tau)} d\tau + 3x(t)$$

$$= \cancel{u(t)} \int_0^t = \cancel{u(t)} e^{-t} \int_0^t x(\tau) e^{\tau} d\tau + 3x(t)$$

$$\mathcal{F}\left\{\frac{dy(t)}{dt} + 10y(t)\right\} = \mathcal{F}\left\{\int_{-\infty}^{\infty} x(\tau) * z(t) - x(t)\right\}$$

$$j\omega Y(j\omega) + 10Y(j\omega) = X(j\omega)Z(j\omega) - X(j\omega) = X(j\omega) \left(\frac{1}{1+j\omega} + 2\right)$$

$$\cancel{H(j\omega)} = \frac{1+j\omega}{1+j\omega+2} = \frac{(1+j\omega)(1+j\omega)}{3+2j\omega} = \cancel{\frac{1+j\omega}{3+2j\omega}}$$

$$\cancel{H(j\omega)} = \frac{\frac{1}{1+j\omega} + 2}{10+j\omega} = \frac{2(j\omega)+3}{(10+j\omega)(1+j\omega)} = \cancel{\frac{1}{3+2j\omega}} \frac{\frac{17}{9}}{10+j\omega} + \frac{\frac{1}{9}}{1+j\omega}$$

$$(b) h(t) = \left(\frac{17}{9}e^{-10t} + \frac{1}{9}e^{-t}\right)u(t)$$

$$4.51 (a) \forall x(t), \quad x(t) * h(t) * g(t) = x(t) \Leftrightarrow h(t) * g(t) = \delta(t)$$

$$\Rightarrow \cancel{X(j\omega)H(j\omega)} \Rightarrow H(j\omega)G(j\omega) = 1$$

$$(b) y(t) = \begin{cases} t, & 0 < t < 1 \\ 0, & t < 0 \text{ or } t > 1 \end{cases}$$

$$Y(j\omega) = \int_{-\infty}^{\infty} y(t) e^{-j\omega t} dt = \int_0^1 t e^{-j\omega t} dt = -\frac{1}{j\omega} \left(t e^{-j\omega t} \Big|_0^1 - \int_0^1 e^{-j\omega t} dt \right) = -\frac{1}{j\omega} \left(e^{-j\omega} + \frac{1}{j\omega} (e^{-j\omega} - 1) \right)$$

$$X(j\omega) = \begin{cases} -\frac{1}{j\omega} \left[\left(1 + \frac{1}{j\omega}\right) e^{-j\omega} - \frac{1}{j\omega} \right], & 2 < |\omega| < 3 \\ \text{not exist}, & \text{otherwise} \end{cases}$$

不存在這樣的 $x(t)$

$$(d) h(t) = 2\delta(t) + u_1(t)$$

$$H(j\omega) = 2 + j\omega = \frac{1}{G(j\omega)} \Rightarrow G(j\omega) = \frac{1}{2+j\omega} \Leftrightarrow g(t) = e^{-2t}u(t)$$

$$y(t) = \frac{dx(t)}{dt} + 2x(t)$$

$$(e) H(j\omega) = \frac{(j\omega)^2 + 3(j\omega) + 2}{(j\omega)^2 + 6(j\omega) + 9} = \frac{(j\omega+1)(j\omega+2)}{(j\omega+3)^2} = 1 - \frac{3(j\omega)+7}{(j\omega+3)^2} = 1 - \left(\frac{3}{j\omega+3} + \frac{-2}{(j\omega+3)^2} \right)$$

$$G(j\omega) = \frac{(j\omega+3)^2}{(j\omega+1)(j\omega+2)} = 1 + \frac{3(j\omega)+7}{(j\omega+1)(j\omega+2)} = 1 + \frac{4}{j\omega+1} + \frac{-1}{j\omega+2}$$

$$h(t) = \delta(t) - (3e^{-3t} - 2te^{-3t})u(t) = \delta(t) - (3-2t)e^{-3t}u(t)$$

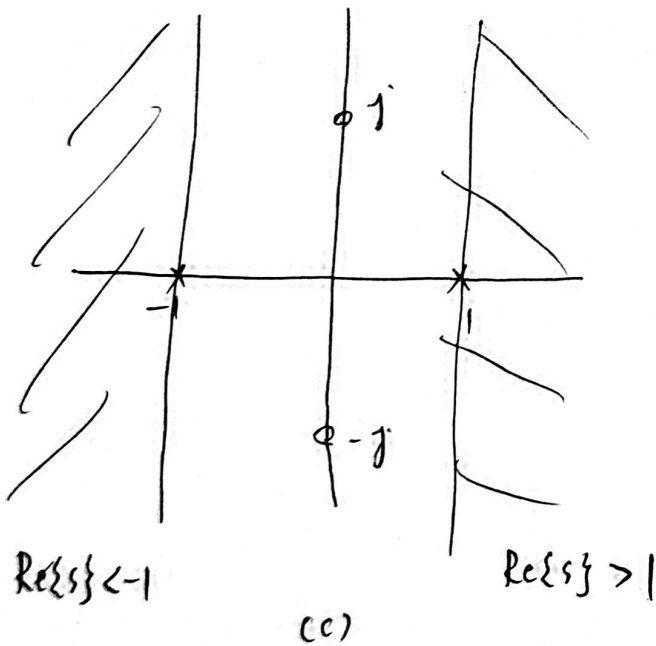
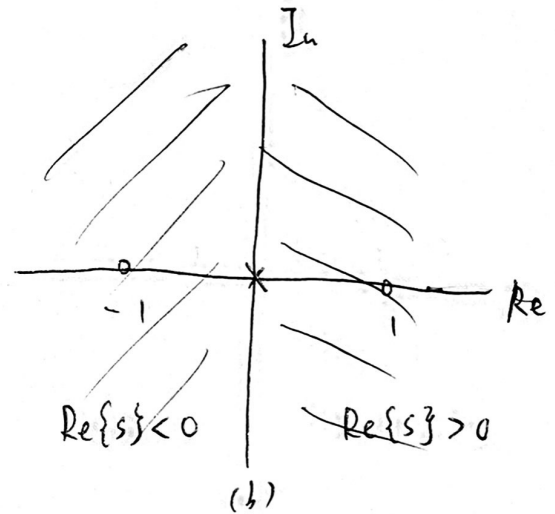
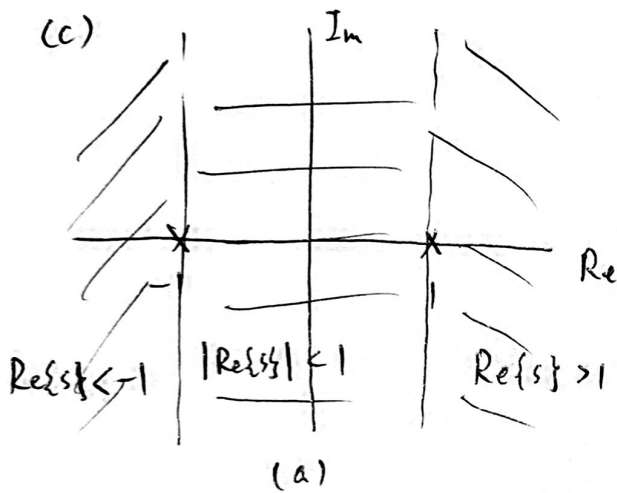
$$g(t) = \delta(t) + (4e^{-t} - e^{-2t})u(t) \quad \text{for the inverse system} \quad \frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y = \frac{dx}{dt} + 6\frac{dx}{dt} + 9x$$

9.41 (a)
$$X(-s) = \int_{-\infty}^{+\infty} x(t) e^{st} dt = - \int_{-\infty}^{+\infty} x(-t) e^{s(-t)} d(-t)$$

$$= \int_{-\infty}^{+\infty} x(t) e^{-st} dt = X(s)$$

(b)
$$X(-s) = \int_{-\infty}^{+\infty} x(t) e^{st} dt = - \int_{-\infty}^{+\infty} x(-t) e^{s(-t)} d(-t)$$

$$= - \int_{-\infty}^{+\infty} x(t) e^{-st} dt = -X(s)$$



Not Even Function

(d)